

Master's Comprehensive Examination

Part I

Theory (90 minutes)

December 6, 2003

Instructions: Answer all four questions. Open book and open notes.

1. At faculty meetings we get pizza. The types are plain, veggie, and pepperoni. We get plain twice as often as the others. The probability of getting pepperoni is the same as veggie.

(a) If 3 pizzas are ordered at random, what is the probability of getting one of each kind?

(b) If 4 are ordered, what is the probability we get at least 1 of each?

(c) If 4 are ordered, what is the probability they are all plain?

2. Let X_1, X_2, X_3 be *independent* Poisson variables with parameters 1, 2, 3, respectively.

(a) Find $P\{X_1 = 1, X_2 = 1, X_3 = 1\}$.

(b) Find the covariance of $Y_1 = X_1 + X_2$ and $Y_2 = X_2 + X_3$.

3. Let X_1, X_2 be a random sample of size 2 from a Poisson distribution with parameter λ . Test $H_0 : \lambda = 1$ vs $H_1 : \lambda > 1$. Find the UMP test of size $\alpha = .0526$. (Note that $X_1 + X_2 \sim P(2\lambda)$).

4. Let X_1, X_2 be independent normal variables with means (μ, μ) , respectively and with variances 1. For estimating μ :

(a) Find the MSE of $\frac{X_1 + X_2}{2}$.

(b) Find the MSE of $X_2 + 1$.